Airline customer satisfaction: a bayesian prediction model

Guglielmi Leonardo • Marian Sorin Nicu • Parimbelli Matteo

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Abstract

In this project we define and fit a Bayesian prediction model for the satisfaction level of customers of an undisclosed airline, based on a set of flight characteristics. Model parameters are derived using MCMC methods via the JAGS tool. We compare a logistic regression under weakly informative priors against a horseshoe shrinkage prior over 22 predictors, assess both with WAIC and PSIS-LOO, and validate the selected model on a held-out set of 128 274 *customers.*

Contents

1. Problem description and data exploration

1.1 Problem description

The goal of this project is to predict the customer satisfaction level of an unspecified airline, given a set of covariates X that describe several characteristics of the flight, such as service-quality items, type of customer, departure delay and so on, and customer's opinion on the flight, like the rating given to some service quality items. Given the high degree of uncertainty inherent in personal opinions, we chose to adopt a Bayesian logistic regression model[1]. The Bayesian coefficients are estimated via Markov Chain Monte Carlo (MCMC) methods.

1.2 Dataset description

The dataset [2] contains 129,881 customer records, described by 22 columns, organised as follows:

- **Target variable**: represents the satisfaction level of customer i, denoted y_i, which can be either *dissatisfied* or *satisfied* and is encoded respectively as $\{0,1\}$.
- **Categorical variables (3)**: indicate the category of the customer and of the flight to which the record refers. For these variables, one-hot encoding is used.
  Variable list: `Customer type, Type of Travel, Class`.
- **Discrete variables (15)**: mostly composed of quality-of-service variables.
  Variable list (service-quality items excluded): `Age`.
  - **Service-quality variables (14)**: indicate the various levels of service available throughout the whole journey, from booking to arrival. These variables contain several zeros, which may indicate that the corresponding service was not available during the flight.
    Variable list: `Seat comfort, Departure/Arrival time convenience, Food and drink, Gate location, Inflight wifi service, Inflight entertainment, Online support, Ease of Online booking, On-board service, Leg room service, Baggage handling, Check-in service, Cleanliness, Online boarding`.
- **Flight properties (3)**: report the spatial and temporal properties of the flight. They are treated as continuous variables.
  Variable list: `Flight distance, Departure delay, Arrival delay`.
- **Modified variables**:
  - owing to the extremely high correlation between `Departure delay` and `Arrival delay`, we decided to replace the latter with the difference between the two delays, so as to retain the information while reducing the correlation.
  - given the high skewness of variables `Departure Delay` and `Mid-flight delay`, we decided to correct it trough logarithm transormation. `Flight Distance` was tested but kept on its original scale as transforming it worsened rather than improving the prediction.

Further details on the procedures are given in Appendix A.

1.3 Dataset exploration

We begin our exploration by examining the balance of the target variable for the prediction model. As shown in Fig. 1, the two classes are fairly balanced, so the use of models more complex than a Bernoulli probability distribution is not justified and will not be pursued.

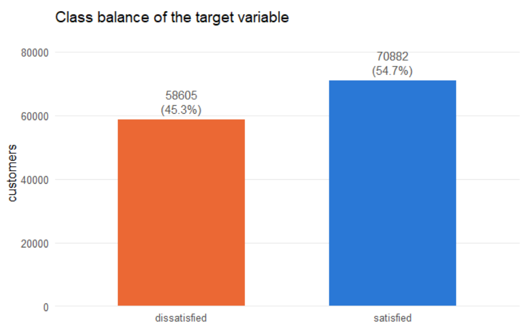
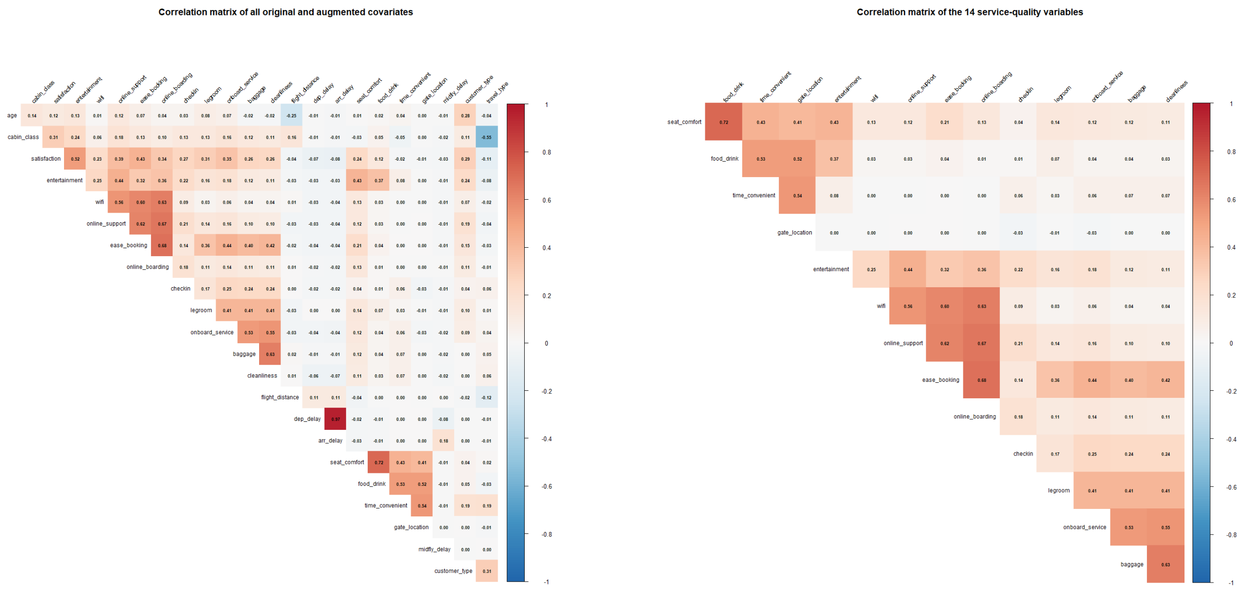


Figure 1. Target variable class balance



(a) Correlation among all variables, both original and augmented

(b) Correlation among service-quality items

To obtain a first indication of the relationship between the response variable and the remaining covariates, we examine their correlation, as shown in Fig. 2a. Focusing on the service-quality variables (Fig. 2b), we can distinguish three subgroups of features. The correlation is computed using R's `corr()` function, which is based on Pearson's correlation coefficient

$$r_{X,Y} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}$$

Tab. 1 reports the class balance of the remaining categorical variables. Given the imbalance among their levels, we decided to stratify the train–test split of the dataset.

Table 1. Class balance of the categorical covariates

Variable	Level	Proportion
Satisfaction	Dissatisfied	0.453
	Satisfied	0.547
Customer type	Disloyal customer	0.183
	Loyal customer	0.817
Travel type	Business travel	0.691
	Personal travel	0.309
Cabin class	Eco	0.449
	Eco Plus	0.072
	Business	0.479

2. Logit model

In this chapter we present the results obtained by fitting a simple Bayesian logistic regression on a reduced set of covariates. The aim is twofold: to establish a first, fully interpretable baseline for the satisfaction prediction problem introduced in Section 1, and to settle a few modelling choices (prior specification, encoding of the cabin class, MCMC setup) that will be reused by the richer shrinkage model of Section 3.

2.1 Covariates and model specification

Following the suggestions of the brief, we fit the model on a compact subset of covariates, using the transformations motivated in the data-exploration phase (Section 1.3): `age`, `seat_comfort`, `flight_distance`, `cabin_class`, the log-transformed departure delay $\log(1 + \texttt{dep\_delay})$ and the signed-log mid-flight delay, together with the two binary descriptors `travel_type` and `customer_type`. All continuous covariates are standardised (zero mean, unit variance) so that coefficients are expressed per standard deviation and the common prior variance is equally informative for every covariate. For the binary covariates the reference level is chosen as the majority group (*Business travel* and *Loyal customer*), so that each coefficient measures the effect of the minority condition relative to the typical customer.

Let $y_i \in \{0,1\}$ be the satisfaction outcome of customer i (encoded as in Section 1) and x_i the corresponding row of the standardised design matrix. We assume a Bernoulli likelihood with a logit link:

$$y_i \mid p_i \sim \text{Bernoulli}(p_i), \qquad \text{logit}(p_i) = \beta_0 + x_i^\top \beta + \beta_{\text{mid}}\,\texttt{midflight\_delay}_i. \tag{1}$$

The mid-flight delay is deliberately kept outside the main design matrix and given its own coefficient β_{mid}, paired with an explicit per-row sampling statement $\texttt{midflight\_delay}_i \sim \mathcal{N}(0,1)$. This treats the (standardised) mid-flight delay as a stochastic node rather than a fixed regressor, which is convenient for handling it as an augmented quantity and keeps its contribution cleanly separated from the other effects.

Prior specification. Since we have no strong prior belief on the magnitude or the sign of the coefficients, and because standardisation already puts every covariate on a comparable scale, we adopt independent, weakly informative Gaussian priors on all the parameters:

$$\beta_0 \sim \mathcal{N}(0,100), \qquad \beta_j \sim \mathcal{N}(0,100) \ \ (j = 1,\dots,P), \qquad \beta_{\text{mid}} \sim \mathcal{N}(0,100). \tag{2}$$

Here the second argument denotes the prior *variance* $\sigma^2 = 100$ (equivalently a precision $\tau = 1/\sigma^2 = 0.01$ in the JAGS parametrisation). On the logit scale a standard deviation of 10 is very diffuse: it places essentially no informative constraint on plausible coefficient values while still being proper, so the posterior is driven by the likelihood, and the resulting estimates remain directly comparable to a maximum-likelihood fit. This choice reflects our lack of prior information and keeps the baseline model as neutral as possible.

2.2 Encoding of the cabin class

Before committing to the model we address a small preliminary question: is it preferable to encode `cabin_class` as a pair of dummy variables (*Eco Plus* and *Business*, with *Eco* as reference) or as a single centred ordinal score $(1/2/3)$? To decide, we fit the model (1) under both encodings and compare them with the Widely Applicable Information Criterion (WAIC), computed from the pointwise log-likelihood of the MCMC draws.

The dummy encoding attains the lower WAIC (1373.2 against 1379.6 for the ordinal one), i.e. a slightly better expected predictive fit. The difference in expected log predictive density is $\Delta\text{elpd} = -3.2$ with a standard error of 2.7, so the preference, although real, is smaller than two standard errors and therefore modest. Because `cabin_class` is genuinely *nominal* rather than an equally-spaced ordinal scale, and because the dummy specification is both more flexible and lets us report a separate, directly interpretable odds ratio for each class, we adopt the dummy encoding for the rest of the analysis.

2.3 MCMC setup and convergence

Posterior inference is carried out via MCMC in JAGS. We run 3 parallel chains, with an adaptation phase of 1500 iterations, a burn-in of 1000 iterations, 3000 monitored iterations per chain and a thinning factor of 7 to reduce the residual autocorrelation between consecutive draws. Table 2 summarises these choices.

Table 2. MCMC settings used for the logit model

Setting	Value
Chains	3
Adaptation	1500
Burn-in	1000
Iterations (kept)	3000
Thinning	7

Convergence was satisfactory for all parameters. The Gelman–Rubin potential scale reduction factors are essentially 1 (point estimates 1.00–1.01, upper confidence limits up to 1.02, multivariate PSRF = 1.01), the trace plots show well-mixed, stationary chains overlapping around a common region, and the autocorrelation decays quickly to zero within a few lags after thinning. The effective sample sizes range from about 590 (for the intercept and the personal-travel coefficient) to roughly 1350, which is more than adequate for stable posterior summaries. A complete convergence assessment (trace plots, autocorrelation functions and the full diagnostic tables) is deferred to Appendix C.

2.4 Posterior results

Figures for the posterior densities of the coefficients and the odds-ratio forest plot are collected below; the corresponding numerical summaries are reported in Table 3. For each coefficient we report the posterior mean, the 95% credible interval on the log-odds scale (β) and the same quantities transformed to the odds-ratio scale ($\mathrm{OR} = e^{\beta}$), which is the natural scale for interpretation.

Table 3. Posterior summaries of the logit coefficients and their odds ratios, sorted by decreasing OR. CIs are 95% credible intervals.

Term	$\bar{\beta}$	$\beta_{2.5\%}$	$\beta_{97.5\%}$	OR	$\mathrm{OR}_{2.5\%}$	$\mathrm{OR}_{97.5\%}$
Class: Business (vs Eco)	1.450	1.120	1.804	4.265	3.065	6.075
Seat comfort	0.632	0.485	0.768	1.881	1.625	2.156
Class: Eco Plus (vs Eco)	0.050	−0.441	0.524	1.052	0.643	1.688
Age	−0.026	−0.164	0.127	0.975	0.848	1.135
Arrival delay residual (signed-log)	−0.041	−0.187	0.111	0.960	0.830	1.118
Flight distance	−0.052	−0.185	0.090	0.949	0.831	1.094
Intercept	−0.105	−0.400	0.180	0.901	0.670	1.198
Personal Travel (vs Business travel)	−0.168	−0.498	0.166	0.846	0.608	1.181
$\log(1+\text{Departure Delay})$	−0.230	−0.371	−0.078	0.795	0.690	0.925
disloyal Customer (vs Loyal)	−1.825	−2.253	−1.431	0.161	0.105	0.239

2.5 Interpretation

Four effects clearly separate from the "no-effect" line (OR = 1). By far the largest is the cabin class: travelling in *Business* rather than *Eco* multiplies the odds of being satisfied by roughly 4.3 (95% CI $[3.07, 6.08]$), even after adjusting for all the other covariates, confirming the marginal contrast already seen in the data-exploration phase. *Seat comfort* is the strongest continuous driver (OR $\approx$ 1.88 per standard deviation, $[1.63, 2.16]$): more comfortable seating is strongly associated with satisfaction. In the opposite direction, a *disloyal customer* has markedly lower odds of being satisfied (OR $\approx$ 0.16, $[0.11, 0.24]$), and a longer departure delay reduces them too (OR $\approx$ 0.80 per unit of $\log(1+\text{delay})$, $[0.69, 0.93]$).

The remaining covariates have credible intervals that comfortably include 1 and cannot be called confident drivers on their own: *Eco Plus* is statistically indistinguishable from *Eco*, while *age*, *flight distance* and the *mid-flight delay residual* all show weak, uncertain effects.

The most instructive case is the *type of travel*. In the exploratory analysis personal-travel customers appeared markedly less satisfied, an almost certain marginal association; here, once cabin class and the other covariates are held fixed, the credible interval for *Personal Travel* (0.61 to 1.18) comes close to, but no longer excludes, 1. This is exactly the confounding anticipated in Section 1.3: business travellers are disproportionately booked in Business class, so once class enters the model directly it absorbs most of the apparent travel-type effect. Both the borderline effects (flight distance, departure delay) and this partially-explained travel-type signal motivate the richer specification of Section 3, where the full battery of 14 service-quality ratings is added under a shrinkage prior to clarify which of these effects survive alongside the stronger, correlated service signals.

3. Horseshoe shrinkage model

3.1 Motivation

The logit model of Section 2 was fit on the small set of covariates suggested by the brief, taking their usefulness for granted. That is a reasonable baseline, but it leaves open the central question of the analysis: among *all* the available predictors, in particular the fourteen service-quality ratings that were set aside earlier, which ones actually carry signal for customer satisfaction, and which are redundant? Rather than answering this by hand, adding and removing covariates and guessing at their relevance, we prefer a principled, automatic mechanism. We therefore fit a Bayesian logistic regression on the *full* set of covariates under a **horseshoe** shrinkage prior [1], which is designed to pull the coefficients of noise predictors strongly towards zero while leaving genuine signals almost untouched. This lets the data, rather than our prior guesses, single out the meaningful covariates.

3.2 Model specification

We include all 21 standardised predictors: the 14 service-quality items plus `age`, `flight_distance`, the two cabin-class dummies (*Eco Plus*, *Business*), $\log(1+\texttt{dep\_delay})$, *Personal Travel* and *disloyal Customer* – and, as in Section 2, the mid-flight delay is kept outside the design matrix with its own coefficient β_{arr}. The likelihood is again Bernoulli with a logit link:

$$y_i \mid p_i \sim \text{Bernoulli}(p_i), \qquad \text{logit}(p_i) = \beta_0 + x_i^\top \beta + \beta_{\text{arr}}\, \texttt{arr\_delay}_i. \tag{3}$$

The horseshoe prior replaces the common vague Gaussian of Section 2 with a scale-mixture of normals: each coefficient gets its own *local* scale λ_j, and all local scales share a single *global* scale τ,

$$\beta_j \mid \lambda_j, \tau \sim \mathcal{N}(0, \lambda_j^2 \tau^2), \qquad \lambda_j \sim \mathscr{C}^+(0,1), \qquad \tau \sim \mathscr{C}^+(0,1), \tag{4}$$

where $\mathscr{C}^+(0,1)$ denotes the standard half-Cauchy distribution. The global scale τ pulls *all* coefficients towards zero, enforcing overall sparsity, while the heavy Cauchy tails of the local scales λ_j let individual coefficients escape that shrinkage when the data demand it – this is precisely the "a few large, most near zero" behaviour that makes the prior suited to feature selection. The arrival-delay coefficient β_{arr} is given its own local scale λ_{arr} under the *same* global τ, so that it competes for signal on an equal footing with the other 21 predictors. Following the JAGS notes, each half-Cauchy is implemented as a Student-*t* with one degree of freedom truncated to the positive reals, `dt(0, 1, 1)T(0, )`. The intercept keeps a diffuse Gaussian prior, $\beta_0 \sim \mathcal{N}(0,100)$ (i.e. `dnorm(0, 0.01)`), and is deliberately left out of the shrinkage so that the baseline log-odds is not penalised.

A convenient by-product of the mixture form is the *shrinkage weight*

$$\kappa_j = \frac{1}{1+\lambda_j^2}, \qquad \kappa_j \in (0,1), \tag{5}$$

which measures how much coefficient j is pulled towards zero: κ_j near 0 marks a preserved *signal* (essentially unshrunk), whereas κ_j near 1 marks a coefficient shrunk towards *noise*. We use the posterior distribution of the κ_j as our covariate-relevance summary.

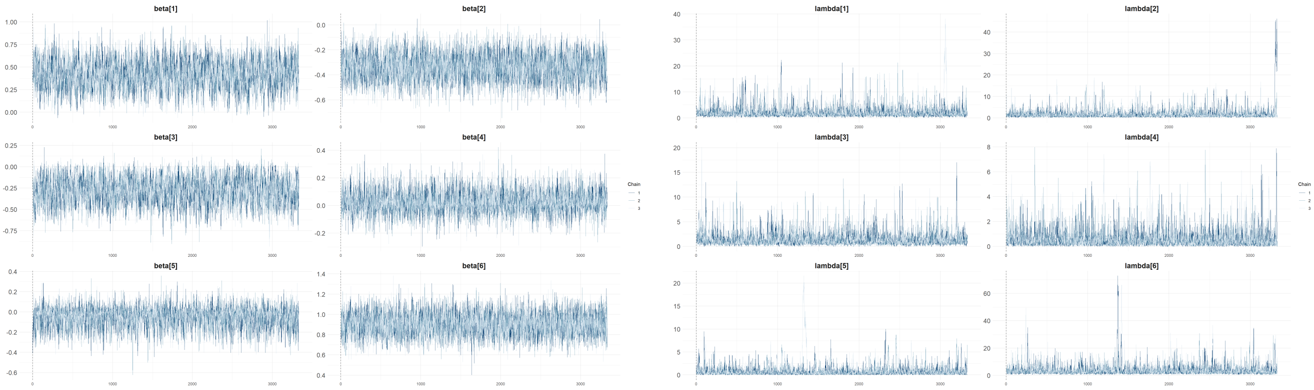
3.3 MCMC setup and convergence

Because of the heavy-tailed half-Cauchy hyperpriors, the horseshoe posterior mixes more slowly than the vague-prior model of Section 2, and it is computationally heavier (22 coefficients, 22 local scales and one global scale, against the 9 parameters of Section 2). We therefore budget a longer sampling run and thinning from the outset; the settings, reported in Table 4, are deliberately modest given the single-core environment and could be extended on a larger machine.

Table 4. MCMC settings used for the horseshoe model

Setting	Value
Chains	3
Adaptation	1500
Burn-in	1000
Iterations (kept)	5000
Thinning	3

Convergence was assessed separately for the two blocks of parameters, the regression coefficients $\{\beta_j, \beta_0, \beta_{\text{arr}}\}$ and the scale parameters $\{\lambda_j, \lambda_{\text{arr}}, \tau\}$, since the latter are the harder to sample. For each block we inspected the Gelman–Rubin potential scale reduction factors, the effective sample sizes, and the trace and autocorrelation plots. The coefficients mix well and show scale-reduction factors close to 1; the scale parameters, and τ in particular, mix more slowly and show higher autocorrelation, as expected for a horseshoe, but reach effective sample sizes adequate for the summaries we report. Figures 3a and 3b show a representative trace plot for each block. The complete diagnostic material – all trace and autocorrelation plots for both the β and the λ parameters, together with the full Gelman–Rubin and effective-sample-size tables – is collected in Appendix C.



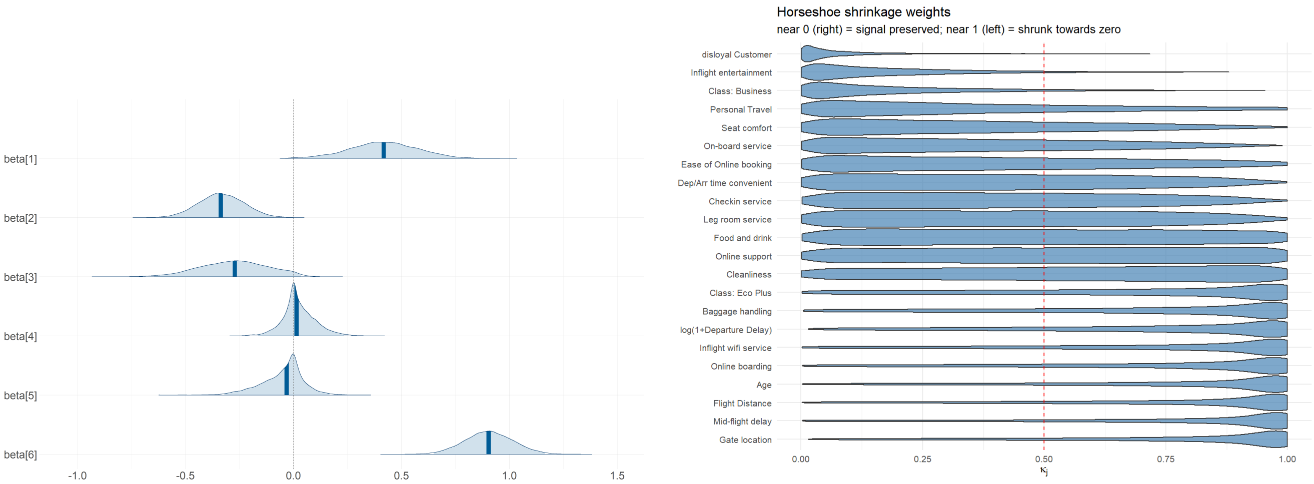
(a) A subset of the regression coefficients β_j.

(b) The local scales λ_j and the global scale τ, which mix more slowly than the coefficients.

Figure 3. Representative trace plots under the horseshoe prior, three chains.

3.4 Posterior results

Figure 4a shows the marginal posterior densities of the coefficients: most concentrate tightly around zero, with only a handful clearly displaced, which is the visual signature of the shrinkage at work. The relevance ranking is made explicit in Figure 4b, which plots the posterior distribution of the shrinkage weights κ_j of Eq. (5): predictors on the "signal" side (κ_j small) are the ones the model retains, while those pushed to $\kappa_j \to 1$ are treated as redundant. Figure 5 reports the same information on the coefficient scale, as a forest plot of the posterior means and 95% credible intervals of the β_j, colouring in the covariates identified as signal ($\kappa_j < 0.5$).



(a) Marginal posterior densities (98% intervals). Most coefficients are shrunk tightly around zero; only a few escape.

(b) Shrinkage weights κ_j. Near 0 (right) the signal is preserved, near 1 (left) the coefficient is shrunk; the dashed line marks $\kappa_j = 0.5$.

Figure 4. Coefficients and shrinkage weights under the horseshoe prior.

3.5 Conclusions

The horseshoe cleanly separates the informative predictors from the redundant ones, and several of its findings sharpen – and in one case reverse – the picture from Sections 1.3 and 2.

The booking / connectivity cluster visibly competes. Among the four mutually correlated "booking / connectivity" items identified in the exploratory analysis (Section 1.3), the prior does not treat them symmetrically: *Ease of Online booking* and *Online support* are preserved noticeably more than *Online boarding* and *Inflight wifi service* – the latter being the single most shrunk item in the whole model. This is exactly the mechanism the horseshoe was chosen for: when several correlated items carry overlapping information, the prior lets two of them represent the signal and treats the other two as largely redundant, instead of forcing us to pick one by hand.

Business class shrinks once service quality is in the model. In Section 2, *Business* class alone carried a large, confident odds ratio (about 4). Here, with all 14 service ratings included, its coefficient shrinks markedly and its credible interval crosses zero. The natural reading is *mediation*, not disappearance: Business passengers are more satisfied largely *because*

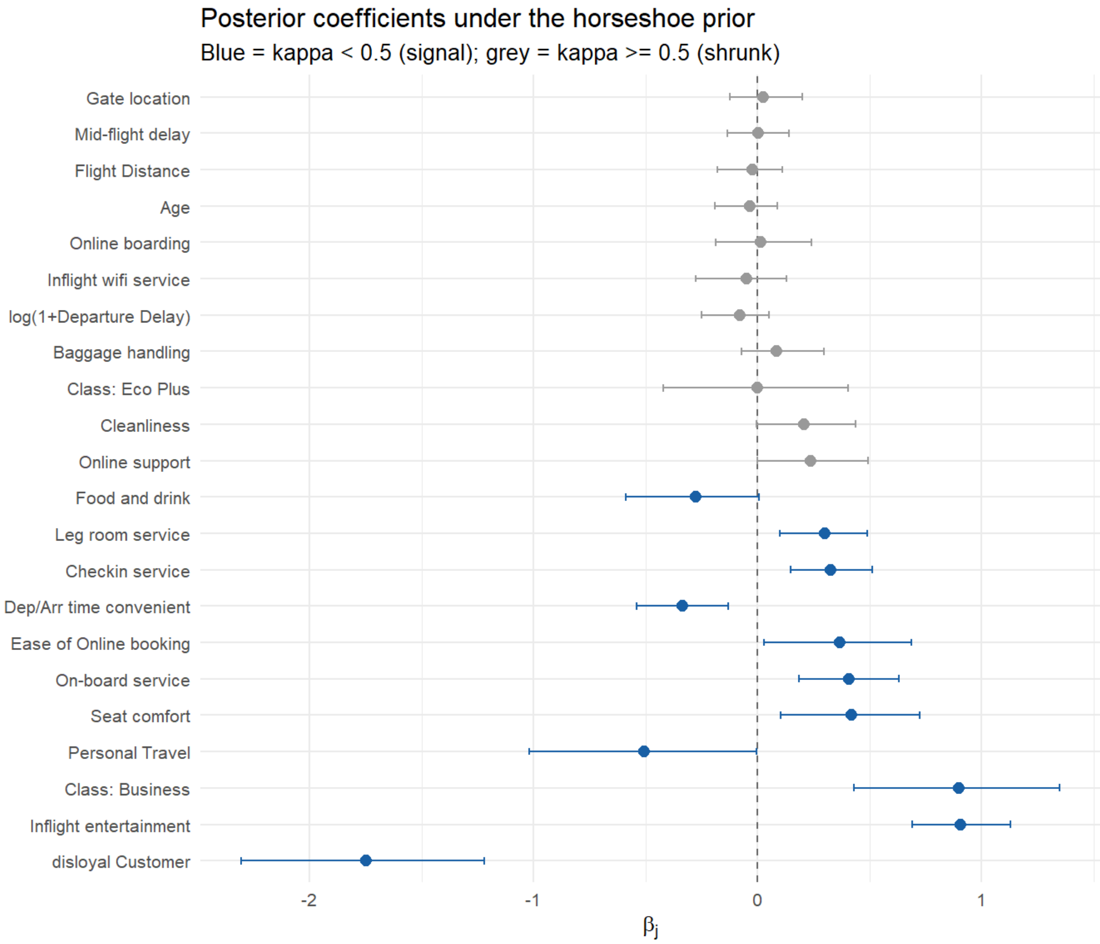


Figure 5. Posterior coefficients β_j under the horseshoe prior (means and 95% credible intervals). Blue marks the covariates identified as signal ($\kappa_j < 0.5$), grey the shrunk ones.

they enjoy better seat comfort, entertainment and service, so in Section 2 the *Class* dummy was partly standing in for those ratings simply because they were not yet in the model.

Personal Travel gets stronger, not weaker. Section 2 found the type-of-travel effect nearly explained away by *Class*. Under the horseshoe, once the service ratings control for the *quality* of the trip experience, *Personal Travel* re-emerges as one of the strongest and most confidently negative effects in the model, with a credible interval lying entirely below zero. The two findings are not contradictory: they show that the earlier attenuation was specifically about *Class* soaking up part of the travel-type signal, not about travel type being unimportant.

A cautionary sign on time convenience. More tentatively, *Departure/Arrival time convenient* has a *negative* posterior mean with a credible interval excluding zero – higher ratings associated with *lower* satisfaction, which is counter-intuitive. Recall from Section 1.3 that this item has one of the largest shares of structural zeros and sits in its own correlated cluster with *Seat comfort*, *Food and drink* and *Gate location*; its zeros plausibly encode “not applicable” rather than genuine dissatisfaction. We therefore read this coefficient as a candidate symptom of that ambiguity rather than a reliable substantive finding, and revisit it directly in the structural-zero robustness check of Section 4.3.

Loyalty dominates throughout. Finally, *disloyal Customer* remains, by a wide margin, the single strongest and most confidently preserved effect in the entire model – consistent across the exploratory analysis and both models of Sections 2 and 3, regardless of how many other covariates are present.

4. Prior Sensitivity and Structural Robustness

This section stress-tests the conclusions of Section 02 (vague-prior logit) and Section 03 (horseshoe) against the modelling choices we made along the way. We ask four questions: does the vague model's posterior depend on the prior *variance* we picked? Does the horseshoe's signal/noise verdict depend on the *global scale* of its half-Cauchy prior? Does the counter-intuitive `Departure/Arrival time convenient` coefficient from Section 03 survive dropping the ambiguous structural-zero rows? And do any of the conclusions depend on the size of the training set?

All of the checks below require heavy JAGS fits, so they are split into separate script files (`04a_prior_sweep.R`, `04b_horseshoe_prior_sweep.R`, `04c_structural_zero_refit.R`) to avoid excessively long cells. Each script is meant to be executed once and its stored output re-used.

4.1 Logit model prior sensitivity

We refit Section 02's model (vague priors, dummy `Class` encoding) four times, varying only the prior variance on every β_j:

$$\sigma_\beta^2 \in \{1, 10, 100, 10000\}.$$

The fits are produced by `04a_prior_sweep.R`. Figure 6 overlays the posterior mean and 95% credible interval of each coefficient across the four prior variances.

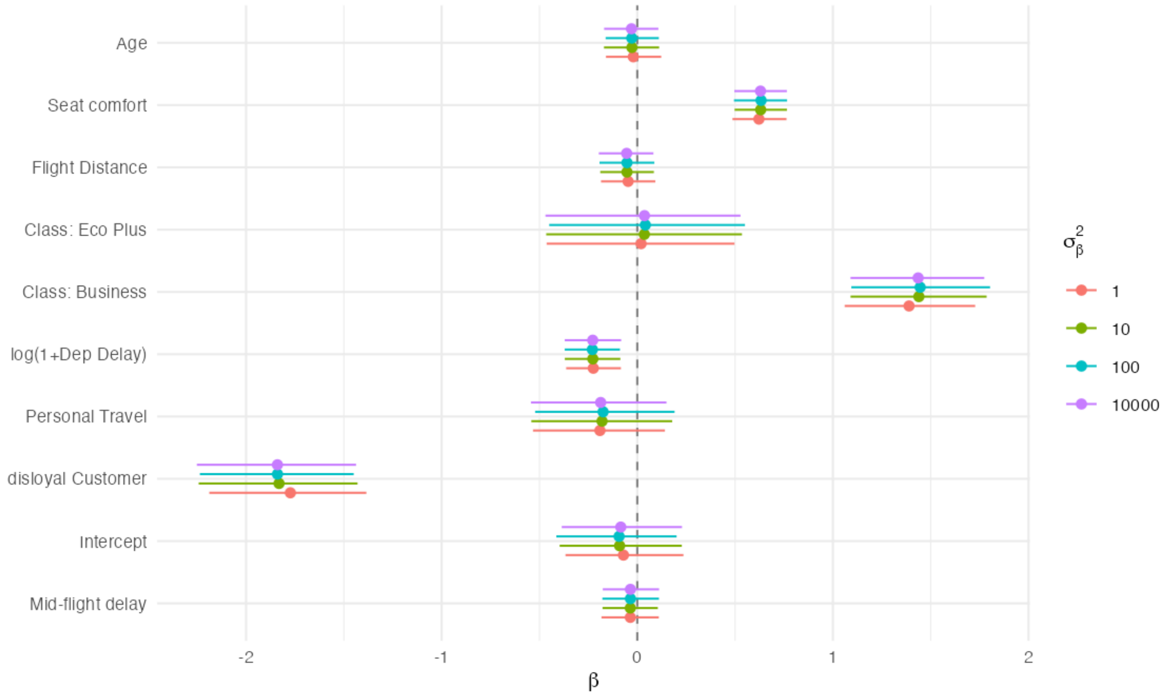


Figure 6. Posterior coefficients of the vague-prior logit model across four prior variances.

Table 5 reports, for each coefficient, the range (max − min) of its posterior mean across the four prior variances.

Table 5. Range of posterior means across the four prior variances (rows are ordered by decreasing range in the notebook).

Term	Range of posterior mean
Intercept	0.022
Age	0.009
Seat comfort	0.011
Flight Distance	0.008
Class: Eco Plus	0.022
Class: Business	0.057
log(1+Departure Delay)	0.005
Personal Travel	0.016
disloyal Customer	0.066
Mid-flight delay (signed-log)	0.001

The largest movement across four orders of magnitude of prior variance is only 0.066 on the log-odds scale, for a coefficient whose posterior mean is itself on the order of 1–2. At $n \approx 1200$ the likelihood therefore dominates all of the tested priors.

4.2 Horseshoe prior sensitivity: does the global scale matter?

The horseshoe's shrinkage behaviour is governed by the scale s of the half-Cauchy prior placed on the **global** scale τ:

$$\tau \sim \mathscr{C}^+(0, s).$$

Small s pulls every coefficient hard towards zero (more aggressive sparsity); large s relaxes the global shrinkage. This is the parameter whose choice could plausibly change *which* covariates the model keeps, so it is the one worth testing.

`04b_horseshoe_prior_sweep.R` refits the exact Section 03 model three times, varying only this global scale, $s \in \{0.1, 1, 10\}$ ($s = 1$ is the Section 03 default), while the local scales λ_j keep the standard $\mathscr{C}^+(0,1)$. For each fit it stores the posterior mean and 95% credible interval of every coefficient, and the per-predictor shrinkage weight

$$\kappa_j = \frac{1}{1+\lambda_j^2}$$

(near 0 = signal, near 1 = shrunk, exactly as in Section 03).

4.2.1 Coefficients across global scales

Figure 7 overlays the posterior coefficients under the three global scales, and Table 6 lists the eight predictors whose posterior mean moves most across them.

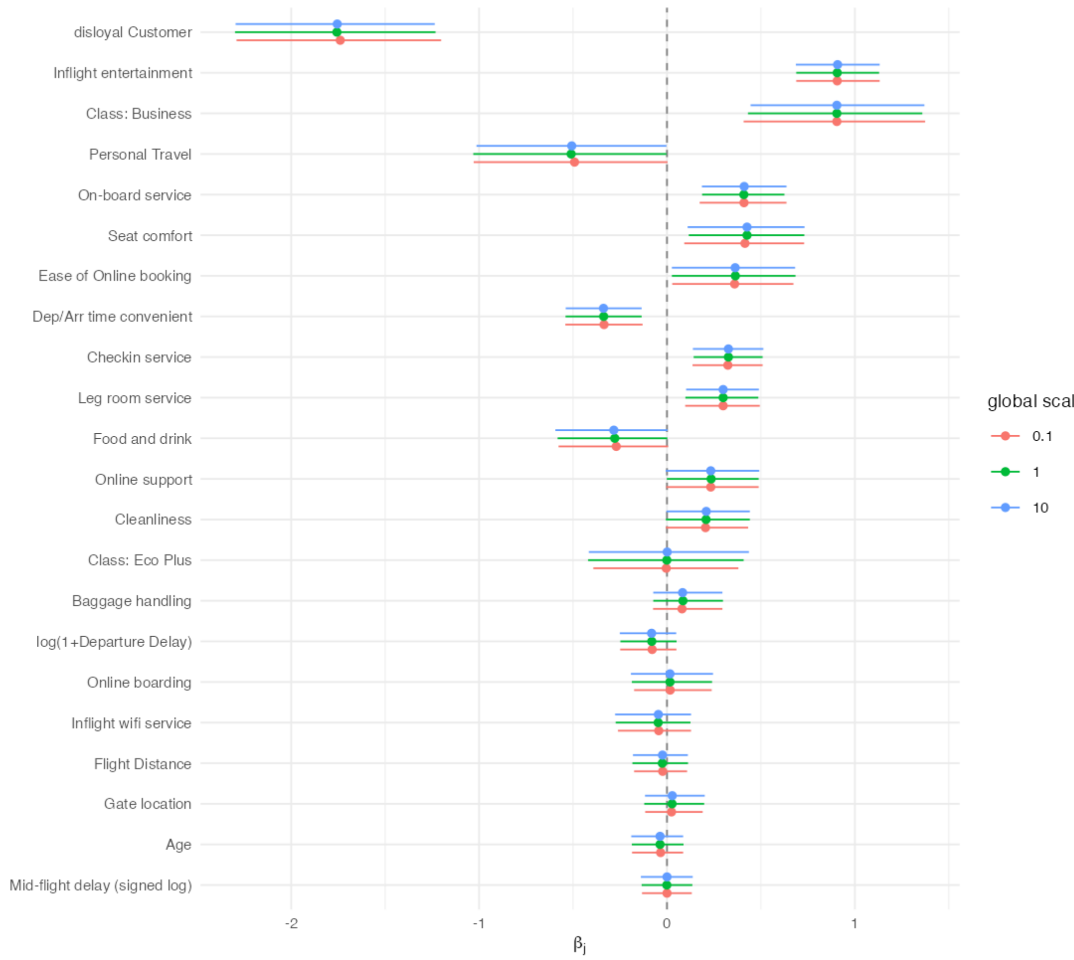


Figure 7. Horseshoe posterior coefficients across three global scales, ordered by shrinkage weight at the default scale.

4.2.2 Shrinkage weights and signal/noise stability

Table 7 gives the posterior shrinkage weight κ_j for every predictor at each global scale, and Table 8 turns each κ_j into a signal/noise verdict ($\kappa_j < 0.5 \Rightarrow$ signal), flagging as `stable` the predictors that keep the same verdict at all three scales.

Across a hundred-fold range of the global scale, one of the 22 predictors changes its signal/noise classification. The covariates the model calls signal at the default scale ($s = 1$) — `disloyal Customer`, `Inflight entertainment`, `Class:`

Table 6. Largest movement of the posterior mean across the three global scales (top eight predictors).

Term	Range of posterior mean
Personal Travel	0.018
disloyal Customer	0.018
Food and drink	0.013
Seat comfort	0.011
Baggage handling	0.005
Class: Eco Plus	0.005
Ease of Online booking	0.004
Cleanliness	0.004

Table 7. Posterior shrinkage weight κ_j at each global scale (one row per predictor; 22 predictors in total).

Term	$s = 0.1$	$s = 1$	$s = 10$
disloyal Customer	0.052	0.071	0.076
Inflight entertainment	0.133	0.171	0.181
Class: Business	0.145	0.180	0.191
On-board service	0.312	0.364	0.378
Personal Travel	0.316	0.349	0.362
Seat comfort	0.326	0.365	0.382
Ease of Online booking	0.361	0.416	0.429
Checkin service	0.370	0.426	0.434
Dep/Arr time convenient	0.371	0.418	0.419
Leg room service	0.397	0.449	0.465
Food and drink	0.449	0.489	0.489
Online support	0.485	0.525	0.534
Cleanliness	0.509	0.544	0.550
Class: Eco Plus	0.653	0.667	0.667
Baggage handling	0.673	0.692	0.703
log(1+Departure Delay)	0.684	0.700	0.717
Inflight wifi service	0.705	0.740	0.739
Online boarding	0.714	0.728	0.732
Age	0.735	0.765	0.766
Flight Distance	0.751	0.752	0.770
Gate location	0.751	0.756	0.750
Mid-flight delay (signed log)	0.770	0.780	0.783

Table 8. Signal ($\kappa_j < 0.5$) at each global scale; `stable` indicates the same verdict across all three scales.

Term	$s = 0.1$	$s = 1$	$s = 10$	stable
disloyal Customer	yes	yes	yes	✓
Inflight entertainment	yes	yes	yes	✓
Class: Business	yes	yes	yes	✓
On-board service	yes	yes	yes	✓
Personal Travel	yes	yes	yes	✓
Seat comfort	yes	yes	yes	✓
Ease of Online booking	yes	yes	yes	✓
Checkin service	yes	yes	yes	✓
Dep/Arr time convenient	yes	yes	yes	✓
Leg room service	yes	yes	yes	✓
Food and drink	yes	yes	yes	✓
Online support	yes	no	no	
Cleanliness	no	no	no	✓
Class: Eco Plus	no	no	no	✓
Baggage handling	no	no	no	✓
log(1+Departure Delay)	no	no	no	✓
Inflight wifi service	no	no	no	✓
Online boarding	no	no	no	✓
Age	no	no	no	✓
Flight Distance	no	no	no	✓
Gate location	no	no	no	✓
Mid-flight delay (signed log)	no	no	no	✓

`Business`, `On-board service`, `Personal Travel`, `Seat comfort`, `Ease of Online booking`, `Checkin service`, `Dep/Arr time convenient`, `Leg room service` and `Food and drink` — are largely preserved across scales, and the largest shift in any posterior mean is 0.018 on the log-odds scale. The strong effects identified in Section 03 (`disloyal Customer`, `Personal Travel`, `Seat comfort`) stay strong and confidently non-zero regardless of s; only the coefficients already sitting near the $\kappa = 0.5$ boundary drift, which is exactly where sensitivity to the global scale is expected and where we drew no firm conclusions in the first place. The headline reading of Section 03 is therefore not an artefact of the particular half-Cauchy scale we chose.

4.3 Structural-zero robustness check

Section 03 speculated that `Departure/Arrival time convenient`'s counter-intuitive *negative* coefficient might be an artefact of the structural zeros flagged in Section 00 (zeros that plausibly mean "not applicable" rather than genuine dissatisfaction). We test that directly by refitting the full horseshoe model with those rows excluded (`04c_structural_zero_refit.R`): 110 rows are dropped, leaving 1103.

Table 9 reports the refit, and Figure 8 compares the "seat/catering/ground" cluster with and without the ambiguous rows.

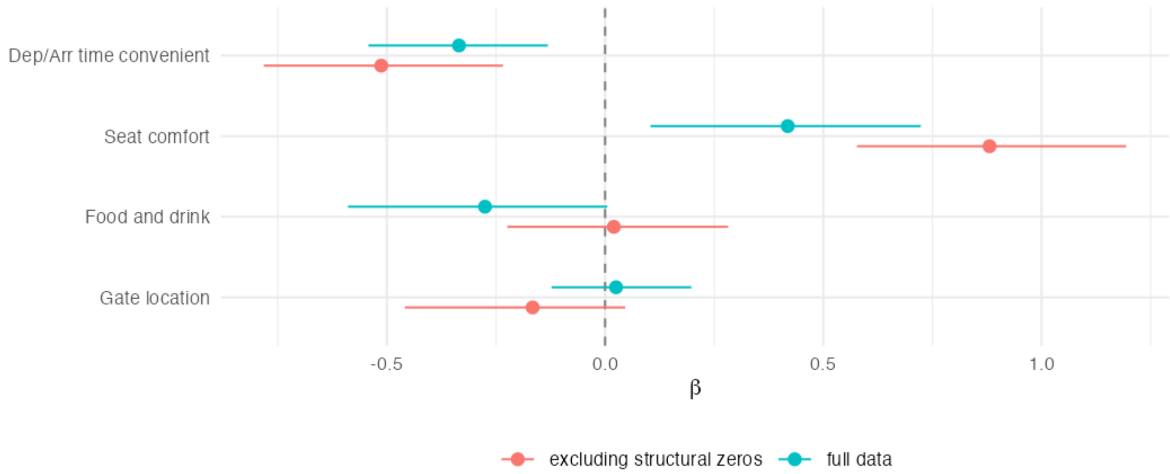


Figure 8. Coefficients of the correlated seat/catering/ground cluster, full data vs. excluding structural-zero rows.

Table 9. Horseshoe refit excluding structural-zero rows.

Term	β mean	β lo	β hi	κ_j
Disloyal Customer	−2.286	−2.975	−1.599	0.059
Inflight entertainment	+1.090	+0.833	+1.361	0.165
Class: Business	+0.976	+0.387	+1.535	0.199
Seat comfort	+0.881	+0.577	+1.194	0.209
Personal Travel	−0.641	−1.278	−0.008	0.331
On-board service	+0.521	+0.231	+0.798	0.338
Dep/Arr time convenient	−0.513	−0.782	−0.234	0.347
Ease of Online booking	+0.493	+0.090	+0.902	0.370
Checkin service	+0.378	+0.154	+0.595	0.420
Leg room service	+0.297	+0.056	+0.529	0.486
Online support	+0.249	−0.021	+0.548	0.548
Gate location	−0.166	−0.459	+0.046	0.618
Cleanliness	+0.158	−0.054	+0.426	0.630
Baggage handling	+0.130	−0.058	+0.386	0.662
Class: Eco Plus	+0.014	−0.467	+0.505	0.670
Inflight wifi service	−0.106	−0.396	+0.103	0.684
log(1+Departure Delay)	−0.087	−0.294	+0.067	0.714
Food and drink	+0.020	−0.224	+0.282	0.733
Online boarding	+0.032	−0.202	+0.293	0.735
Flight Distance	−0.016	−0.194	+0.149	0.763
Age	−0.028	−0.191	+0.114	0.779
Mid-flight delay (signed log)	−0.013	−0.185	+0.144	0.780

The hypothesis does not survive the check. `Departure/Arrival time convenient`'s negative coefficient does not shrink towards zero when the ambiguous rows are removed — it gets *more* negative (from −0.335 to −0.513), while its credible interval widens slightly, as expected with 110 fewer rows. Whatever drives this effect, it is not simply the structural-zero ambiguity we suspected in Section 03; we report this as an open, unresolved finding rather than force a tidier story onto it. On the other hand, the other two members of that same correlated cluster move in *opposite* directions once the ambiguous rows are excluded. `Seat comfort` strengthens substantially (κ drops from 0.367 to 0.209, β rises from +0.419 to +0.881), whereas `Food and drink` goes from a borderline signal ($\kappa = 0.492$, $\beta = -0.275$) to firmly shrunk ($\kappa = 0.733$, $\beta = +0.020$). The structural zeros therefore do not add uniform noise across the cluster; they affect its members differently.

4.4 Sample-size robustness

The logit prior-variance sweep (Section 4.1) and the horseshoe global-scale sweep (Section 4.2) varied the prior; the structural-zero refit (Section 4.3) varied which rows enter the fit. This one varies how many. Every model in this report is estimated on 1213 records, about 1% of the cleaned dataset, a choice forced by MCMC cost. `04d_large_subsample_refit.R` refits the model of Section 2 on a stratified subsample of 15 003 records. This leaves the coefficients we rely on unmoved (`Class: Business` by 0.005, `disloyal Customer` by 0.014, `Seat comfort` by 0.022) and contracts the intervals by a median factor of 3.42, against the 3.52 that $\sqrt{n}$ asymptotics predicts. Three weak effects (`Flight Distance`, `Age`, `Mid-flight delay`) acquire intervals excluding zero, which is a limitation of the $n = 1213$ analysis rather than a finding about the airline.

5. Prediction and validation

Everything up to this point has been estimated on the 1203-customer training subsample. This section confronts the model with the 128 274 held-out customers set aside in Section 1, which were never used for fitting, standardisation or variable selection. Following the conclusion of Section 4, the horseshoe model is the one carried forward here; the corresponding comparison against the vague-prior logit and against BAS is reported in Appendix D.

5.1 From posterior draws to predicted probabilities

For a held-out customer with covariate row $\tilde{x}_i$, the Bayesian point prediction is the posterior mean of the success probability, obtained by pushing every retained draw through the inverse link and averaging:

$$\hat{p}_i = \frac{1}{S}\sum_{s=1}^{S} \operatorname{logit}^{-1}\Big(\beta_0^{(s)} + \tilde{x}_i^\top \beta^{(s)} + \beta_{\text{arr}}^{(s)}\, \texttt{arr\_delay}_i\Big), \tag{6}$$

This averages the transformed draws rather than transforming the averaged ones: $\operatorname{logit}^{-1}$ is non-linear, so $\mathbb{E}[g(\theta)] \neq g(\mathbb{E}[\theta])$, and only the former propagates posterior uncertainty. Test covariates are standardised with the *training* means and standard deviations, so nothing from the held-out set enters the fit.

5.2 Discrimination and probabilistic accuracy

We report two complementary summaries. The area under the ROC curve (AUC) measures how well the model *ranks* customers and is invariant to the cutoff. The Brier score,

$$\text{BS} = \frac{1}{n}\sum_{i=1}^{n}\big(\hat{p}_i - y_i\big)^2, \tag{7}$$

is a strictly proper scoring rule, minimised only when the stated probabilities match the true ones, so it rewards predictions that are both well ranked and numerically honest. On the held-out set the horseshoe model attains an AUC of 0.900 and a Brier score of 0.126.

5.3 Classification and the choice of cutoff

Turning probabilities into hard classifications requires a cutoff. Table 10 contrasts the conventional 0.5 with the training base rate $\bar{y} = 0.546$, the natural alternative when the two classes are not perfectly balanced.

Table 10. Classification performance on the held-out set at two cutoffs.

Cutoff	Sensitivity	Specificity	Accuracy
$t = 0.5$	0.837	0.812	0.826
$t = \bar{y} = 0.546$	0.815	0.843	0.828

Moving to the base rate trades about two points of sensitivity for three of specificity and leaves accuracy essentially unchanged (0.826 against 0.828). That near-indifference is informative: the errors are not concentrated on one side, and accuracy alone is a blunt instrument for choosing an operating point. The right cutoff depends on what the two mistakes cost, which is Section 5.6.

5.4 Calibration

Neither AUC nor accuracy asks whether the predicted probabilities *mean* what they claim: among customers the model calls 70% likely to be satisfied, roughly 70% should be. This matters as much as discrimination here, since the posterior predictive probability is the object we report. Table 11 bins the held-out set into deciles of predicted probability against the observed rate.

Calibration is good in the upper half: from the seventh decile up, predicted and observed agree closely (0.801 against 0.819, 0.896 against 0.937). In the middle the model is mildly over-confident about dissatisfaction, predicting 0.33 and 0.49 in deciles four and five where the observed rates are 0.26 and 0.43, and the lowest two deciles reverse that. So the model separates the classes well at the extremes, where most predictions live, and is least trustworthy around $\hat{p} \approx 0.3$–0.6, where the evidence genuinely is ambiguous.

5.5 Posterior predictive check by subgroup

The checks above compare point predictions against outcomes. A posterior predictive check asks more: taking the fitted model seriously as a generative description and simulating whole replicated datasets from it, do those replications reproduce features of the real data the fit never targeted? For each posterior draw we simulate outcomes for the entire held-out set and recompute the satisfaction rate within each level of `Class`, `Customer Type` and `Type of Travel`, giving a posterior predictive distribution per subgroup to compare the observed rate against.

The observed rates fall inside the bulk of the simulated distributions for every subgroup of all three variables, so the model is not systematically over- or under-predicting satisfaction for any class of customer on data it has never seen. The

Table 11. Calibration by decile of predicted probability.

Decile	Mean predicted	Observed rate	n
1	0.035	0.122	12 828
2	0.100	0.125	12 827
3	0.195	0.163	12 827
4	0.326	0.260	12 828
5	0.489	0.430	12 827
6	0.658	0.637	12 827
7	0.801	0.819	12 828
8	0.896	0.937	12 827
9	0.946	0.984	12 827
10	0.977	0.997	12 828

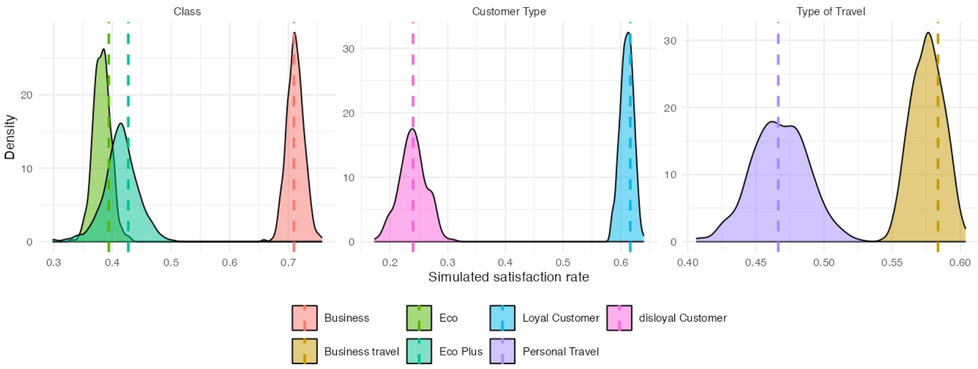


Figure 9. Posterior predictive check by subgroup. Densities are the simulated satisfaction rates; dashed lines mark the rates actually observed in the held-out set.

subgroup rates were never fitted directly, so reproducing them is evidence that the covariate effects were captured rather than merely tuned to the aggregate.

5.6 Choosing the cutoff by expected loss

Rather than pick 0.5 or the ROC-optimal point by convention, we state what the two errors cost and minimise posterior expected loss. The airline flags a customer when $\hat{p}_i < t$ and follows up with a retention contact; write C_{miss} for the cost of missing a dissatisfied customer and C_{waste} for that of contacting a satisfied one. The expected loss at cutoff t is

$$\mathscr{L}(t) \;=\; C_{\text{miss}} \cdot \mathbb{P}\big(\hat{p} \geq t,\; y=0\big) \;+\; C_{\text{waste}} \cdot \mathbb{P}\big(\hat{p} < t,\; y=1\big), \tag{8}$$

estimated on the held-out set. Only the *ratio* of the costs matters, so the airline need not price either error, just say how many wasted contacts it would trade for one retained customer.

Table 12. Cutoff minimising expected loss, by assumed cost ratio $C_{\text{miss}} : C_{\text{waste}}$.

Cost ratio	Optimal cutoff $t^\star$	Expected loss
1 : 1 (equal cost)	0.56	0.172
3 : 1	0.74	0.252
5 : 1	0.80	0.287

Under equal costs the optimum is $t^\star = 0.56$, essentially the base rate and close to the conventional 0.5, a useful sanity check on Equation (8). As missing a dissatisfied customer grows more expensive the cutoff rises, to 0.74 at 3 : 1 and 0.80 at 5 : 1: the airline flags far more customers, accepting wasted contacts to catch the recoverable ones. This is where the Bayesian machinery pays off operationally. Because the model returns a calibrated distribution rather than a label, the same fit supports any cost structure the business specifies, and the cutoff follows from the loss function instead of convention.

6. Conclusions

We modelled airline customer satisfaction with a Bernoulli-logit likelihood, first on the reduced covariate set suggested by the brief and then on the full set under a horseshoe shrinkage prior, with `BAS` model averaging as an independent cross-check. All posteriors were sampled by MCMC in JAGS.

Three effects dominate and survive every specification. Customer loyalty is the strongest by a wide margin and the least shrunk coefficient at every global scale tested, *Business* class follows, and among the fourteen service items *Inflight* *entertainment* is the one the prior preserves most clearly, while *Inflight wifi service* and *Gate location* are heavily shrunk despite sitting in the same correlated clusters. That asymmetry is what the horseshoe was chosen for.

These conclusions are not artefacts of our modelling choices. The vague-prior posterior moves by at most 0.066 across four orders of magnitude of prior variance; one predictor of twenty-two changes its signal/noise verdict across a hundred-fold range of the global scale, and it sits on the $\kappa = 0.5$ boundary; and a training set twelve times larger moves the coefficients we report by at most 0.022. Out of sample, on 128 274 unseen customers, the model reaches an AUC of 0.900, classifies 82.8% correctly, and reproduces the observed satisfaction rate in every subgroup.

One result resisted explanation: *Departure/Arrival time convenient* carries a negative coefficient whose interval excludes zero, and refitting without the structural-zero rows strengthened it rather than weakening it, so we report it as an open question. Three limitations are worth naming. The structural zeros were flagged and tested but not modelled with a zero-inflated likelihood; $n \approx 1200$ is modest for twenty-two predictors, which is why several effects are borderline; and the encoding and link choices of Section 2 were settled on training WAIC without out-of-sample re-validation.

A. Data augmentation

A.1 Decorrelating departure and arrival delays

Inspecting the correlation matrix (Fig. 2a), we notice an extremely high correlation between `Departure delay` and `Arrival delay`. This value makes the two variables almost duplicates of one another. In order to remove this redundancy while retaining as much information as possible, we decided to replace Arrival delay with a quantity that captures the difference between the two.

We considered two options:

- using the plain difference between the two variables;
- fitting a simple linear regression model with `Arrival delay` as the target and `Departure delay` as the covariate, and taking the residual of each observation as the new covariate.

The second model is more general and, in fact, encompasses the first one; however, the first is more interpretable and carries an immediate meaning, such as "mid-flight delay" or "time gained during the flight". The deciding point was whether the second option differed enough from the first to justify its higher complexity.

The regression model that reproduces the plain difference is the one with slope equal to 1 and intercept equal to 0. Fitting the regression model on the full dataset yields an intercept of 0.76 and a slope of 0.97; we therefore decided to opt for the simple difference between the arrival and departure delays.

A.2 Reducing covariates skeweness

From the boxplots in Fig.10 we can see how covariates `Flight Distance`, `Departure Delay` and `Mid-flight delay` are extremelly right-skewed. Since covariate skewnes might slow down MCMC convergence and are prone to present outliers and leverage points, we decided to transform them and check whether the transformation is beneficial.

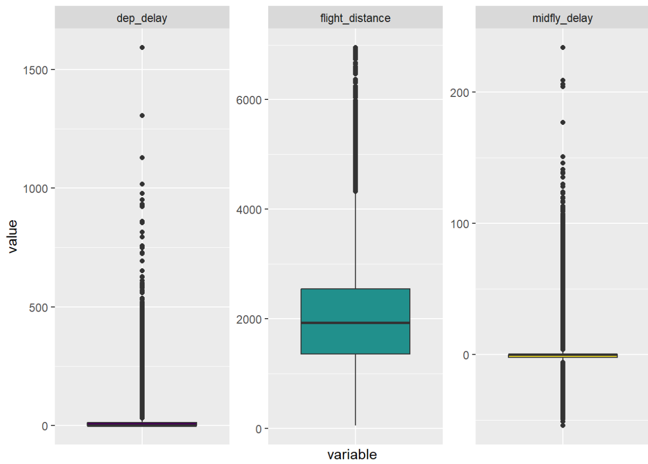


Figure 10. Boxplots of skewed covriates.

Since both `Flight distance` and `Departure delay` are non-negative we transform them using $ln(1+x)$; instead for `Mid-flight delay` we use the *signed-log transformation* $\text{sign}(x) \cdot ln(1+|x|)$ due the presence of negative values.

variable	skewness
Flight Distance	0.466
log1p(Flight Distance)	-1.599
Departure Delay	6.853
log1p(Departure Delay)	0.919
Midflight Delay	2.941
signed-log(Midflight delay)	0.234

Table 13. Skewness of raw and transformed variables

As we can see in Tab.13 `Flight distance` is only midly skewed and the transformation actually turn to be worse than the original varibale, therefore we kept it unchanged. Instead for what concerns `Departure delay` and `Mid-flight delay` we see higher level of skeweness and the transformation improves this aspect; therefore we decided to kee the log-transformed variables.

B. Feature selection cross-check with BAS

After fitting our feature-selection model based on a horseshoe prior, we would like to compare it with a Bayesian Adaptive Sampling (BAS) model in order to verify whether the two methods agree on which covariates are more useful in explaining the target variable.

With 23 predictors, $2^{23} \approx 8.4$ million candidate models are far too many to enumerate exhaustively, so we use `method = "MCMC"` in BAS to stochastically search the model space. A later section assesses whether this search suffers from instability or not.

Table 14 reports the marginal posterior inclusion probabilities $P(\beta_j \neq 0 \mid Y)$ for every predictor, together with the five highest-probability models returned by the search and their summary statistics (Bayes factor, posterior probability, R^2, dimension and log marginal likelihood).

Table 14. Marginal posterior inclusion probabilities and the top five models identified by the BAS MCMC search. A value of 1.000/0.000 in a model column indicates the predictor is included/excluded in that model.

	$P(\beta \neq 0 \mid Y)$	Model 1	Model 2	Model 3	Model 4	Model 5
Intercept	1.000	1.000	1.000	1.000	1.000	1.000
seat_comfort	0.803	1.000	1.000	1.000	1.000	0.000
time_convenient	0.973	1.000	1.000	1.000	1.000	1.000
food_drink	0.500	1.000	0.000	1.000	0.000	0.000
gate_location	0.037	0.000	0.000	0.000	0.000	0.000
wifi	0.049	0.000	0.000	0.000	0.000	0.000
entertainment	1.000	1.000	1.000	1.000	1.000	1.000
online_support	0.220	0.000	0.000	0.000	0.000	0.000
ease_booking	0.954	1.000	1.000	1.000	1.000	1.000
onboard_service	0.996	1.000	1.000	1.000	1.000	1.000
legroom	0.965	1.000	1.000	1.000	1.000	1.000
baggage	0.073	0.000	0.000	0.000	0.000	0.000
checkin	0.986	1.000	1.000	1.000	1.000	1.000
cleanliness	0.310	0.000	0.000	0.000	0.000	0.000
online_boarding	0.029	0.000	0.000	0.000	0.000	0.000
Age_std	0.037	0.000	0.000	0.000	0.000	0.000
FlightDist_std	0.035	0.000	0.000	0.000	0.000	0.000
Class_EcoPlus	0.029	0.000	0.000	0.000	0.000	0.000
Class_Business	0.993	1.000	1.000	1.000	1.000	1.000
logdep_std	0.092	0.000	0.000	0.000	0.000	0.000
TypeTravel_Personal	0.451	0.000	0.000	1.000	1.000	1.000
CustType_disloyal	1.000	1.000	1.000	1.000	1.000	1.000
logmid_delay	0.031	0.000	0.000	0.000	0.000	0.000
BF	NA	1.000	0.574	0.477	0.369	0.268
PostProbs	NA	0.128	0.071	0.059	0.048	0.033
R^2	NA	0.426	0.421	0.430	0.425	0.420
dim	NA	11.000	10.000	12.000	11.000	10.000
logmarg	NA	−516.838	−517.392	−517.578	−517.835	−518.155

C. Parameter-space convergence

We now test the goodness of the MCMC search over the parameter space executed by BAS.

Convergence within one run

BAS reports two independent estimates of each posterior inclusion probability:

- `probne0.RB`: takes the set of distinct visited models that contain variable j, uses their marginal likelihoods, renormalises and sums the mass of those models;
- `probne0.MCMC`: the raw fraction of iterations the chain spent in some model containing variable j.

If the chain has converged, the two estimates should agree closely.

The correlation between the two estimates is 0.9999 and the maximum absolute difference is 0.014: the points sit essentially on the diagonal (Figure 11), the sign that this run has converged.

Stability across seeds

We refit the identical model twice more with different random seeds and compare the posterior inclusion probabilities (PIPs) to check whether they agree. Table 15 lists the predictors with the largest spread, ordered by the range across the three seeds.

BIC prior as a robustness check

We try another prior on the β coefficients: if the search is stable, the effect of the prior should vanish once the number of iterations is large enough. Table 16 compares the g-prior PIPs with those obtained under a BIC-based prior.

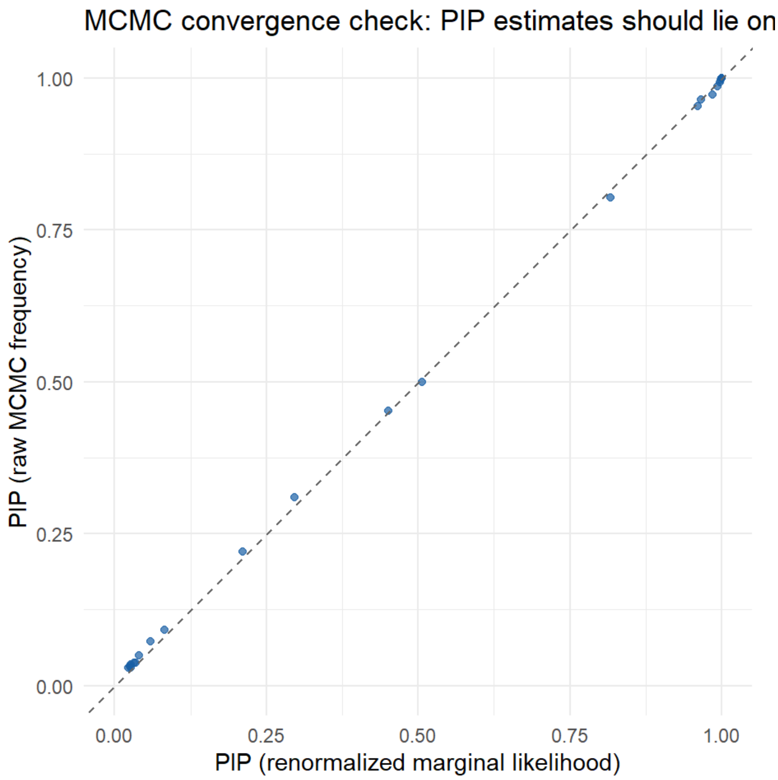


Figure 11. MCMC convergence check. Renormalised-marginal-likelihood inclusion probabilities (x-axis) against raw MCMC-frequency inclusion probabilities (y-axis); points lying on the dashed diagonal indicate the two estimates agree.

Table 15. Posterior inclusion probabilities across three random seeds, with the range (max − min) per predictor. Only the eight most variable predictors are shown.

term	seed123	seed1	seed2	range
seat_comfort	0.803	0.841	0.826	0.038
food_drink	0.500	0.533	0.511	0.033
online_support	0.220	0.228	0.204	0.024
cleanliness	0.310	0.297	0.316	0.019
TypeTravel_Personal	0.451	0.439	0.457	0.017
legroom	0.965	0.951	0.959	0.013
time_convenient	0.973	0.979	0.983	0.011
baggage	0.073	0.065	0.064	0.009

Table 16. Posterior inclusion probabilities under the g-prior and the BIC-based prior, with the absolute difference per predictor (six most affected predictors shown).

term	g_prior	BIC	abs_diff
online_support	0.220	0.224	0.004
cleanliness	0.310	0.314	0.004
seat_comfort	0.803	0.801	0.002
food_drink	0.500	0.498	0.002
ease_booking	0.954	0.952	0.002
legroom	0.965	0.967	0.002

With 100,000 MCMC iterations, PIPs move by at most 0.038 across the three seeds and by at most 0.004 between the g-prior and the BIC-based prior. Given the number of MCMC iterations, the search proves itself to be stable.

D. BAS vs. Horseshoe: model comparison

Finally, we compare the models selected by the two approaches and check on which coefficients they agree and on which they do not. Figure 12 plots each predictor's BAS inclusion probability against the horseshoe's $1-\kappa_j$ shrinkage measure, colouring predictors by whether the two methods make the same signal/noise call.

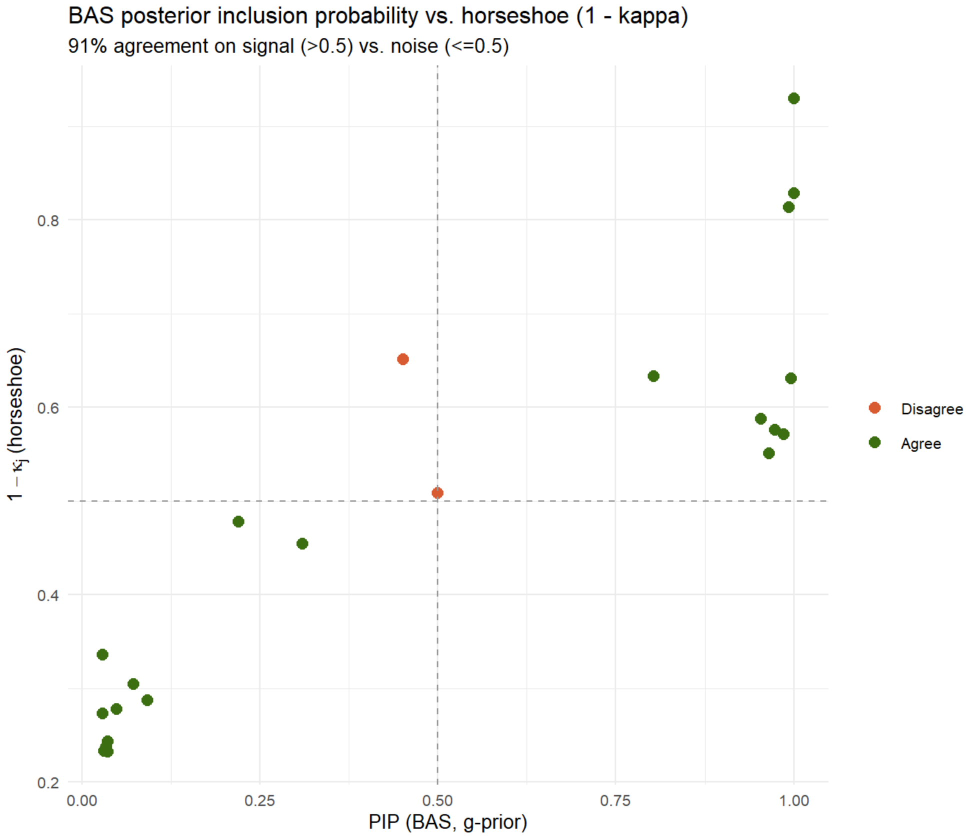


Figure 12. BAS posterior inclusion probability (x-axis) against the horseshoe's $1-\kappa_j$ (y-axis). Dashed lines mark the 0.5 signal/noise cutoff for each method; green points denote agreement, the red point denotes the single disagreement.

91% of the 22 predictors get the same signal/noise call from BAS's model-space search and from the horseshoe's continuous shrinkage — two structurally different estimation strategies (discrete model averaging vs. a single shrinkage prior) landing on the same handful of variables. The one disagreement, `Class: Business` (PIP 0.993, $\kappa = 0.186$), is a genuine borderline case in both methods (BAS just below its 0.5 cutoff, the horseshoe's credible interval for it crossing zero in Section 03) — not a real contradiction between them.

References

[1] M. Villan, *Bayesian Learning Book*. [Online]. Available: https://mattiasvillani.com/BayesianLearningBook/.

[2] *Airline customer satisfaction*. [Online]. Available: https://www.kaggle.com/datasets/raminhuseyn/airline-customer-satisfaction.